

SPECIFICATION

Product Name: IoT Module

Model No.: TI602B2S

Type: URAT Interface

Version: V1.0

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1. Profile

TI602B2S is a module. It consists of a highly integrated chip BL602 and several peripheral components. It supports 2.4Ghz and WIFI standard IEEE 802.11 b/g/n. Adopting a 3.3V single power supply, the module can be flexibly applied to various consumer products. This module application for the smart home, household electrical appliances and smart light control.

1.1 Features

Built-in low power 32bit CPU and also can be an application processor

WIFI standard is 802.11 b/g/n

WPS/WEP/WPA/WPA2 Personal/WPA2 Enterprise/WPA3

PWM control

Support timer operation

Support WIFI remote control

Support timer operation

Supports compatible pairing mode/Bluetooth pairing

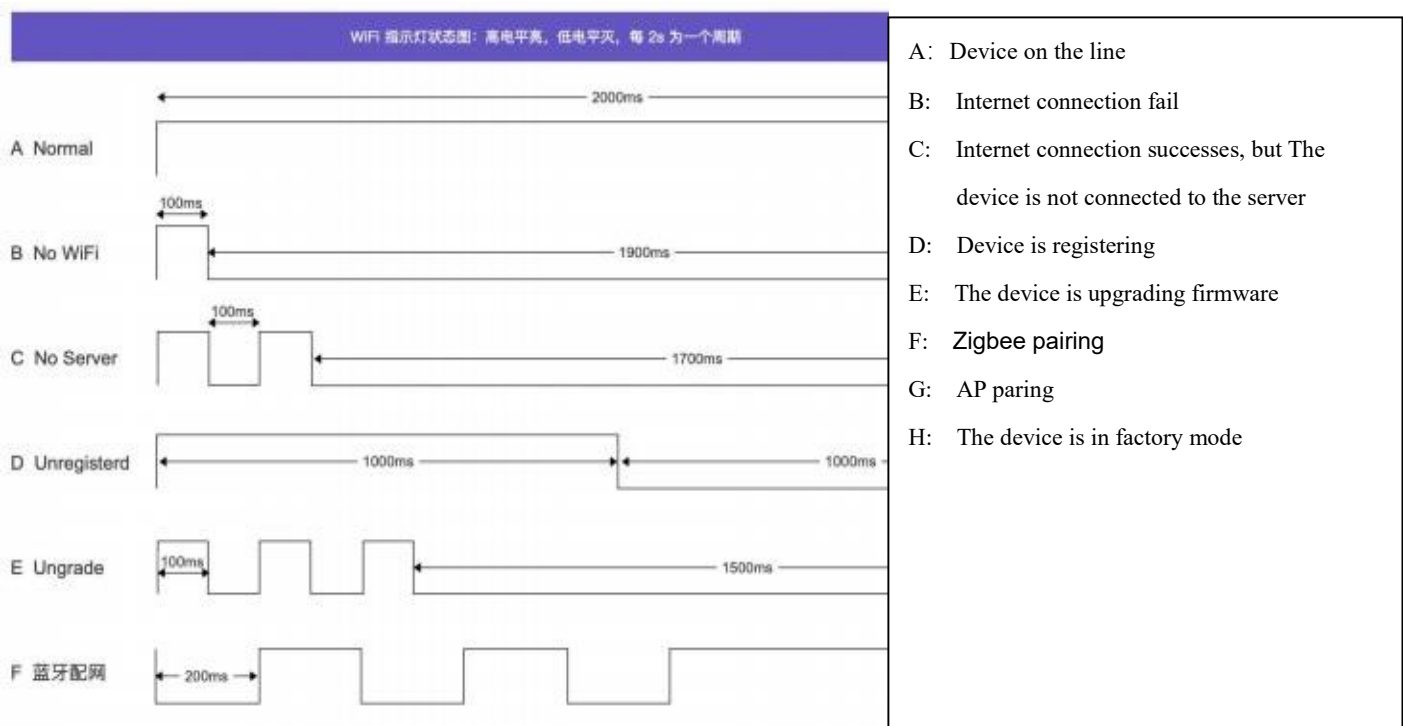
OTA upgrade

LAN control

Support voice control by smart speaker of Alexa, Google assistant , Tmall-genie, Mijia, Xiaodu

2.Function Description

2.1 WIFI indicator light



2.2 Pairing network

TI602B2S has two way for paring network: BLE pairing and compatibility pairing.

2.2 Entering pairing network mode

2.2.1 TI602B2S has some pairing networking mode as following:

1. Unpaired devices automatically enter BLE network mode after powering on
2. Pull down the pin with 5s for any button input when the device not under the pairing network mode, the chipset will enter into BLE pairing mode. Pull down the pin with 5s for any button input again when the device under the BLE pairing mode, the chipset will enter into compatibility pairing.
3. When the App remove the device, the device will be reset as factory default and enter into BLE pairing mode.

2.2.2 Exit the pairing network mode

1. The device will exit the pairing network status when the pairing is successful
2. Over 3mins the device will auto exit pairing network status when the device under the pairing network mode.

2.3 Switch control

Pin of switch input default is High level, effective when the pull down. When triggered, the relay output pin corresponding to the button flips

2.4 Timer control

Support timer type as following:

1. Timing : Set time for the specified action
2. Delay: Execute the specified action after a certain period of time (The delay time max is 23 hours 59mins)

Attention:

Action function only has turn on /off.

Timing and delay have option for enable and disable

Module support 20 timers, but only support 8 timers run at the same time.

2.5 Electrification reaction

Electrification reaction: The SM-028M is powering on again after power off, the module can be set as three mode as following:

- Star
- Close

- Restore the state before the power outage

Factory default as close

2.6 LAN (local area network) Control

App can control the module switch directly when the mobile phone and device are connected under the same local area network without using the cloud platform.

2.7 The indicator light of network

TI602B2S supports configuring network indicator lights through the APP, (light off status when the device online)

2.8 Delete device

User can delete the device by APP, the device remove from the APP, the device auto reset as factory default mode.

2.9 RF433 learning

2.9.1 RF433 learning remote controller

The max learning RF433 control is 10 buttons, the oldest controller will be cover with the newest one when the controller is over 10 pcs.

2.9.2 Entering learning mode

When the device is power on and not under the learning mode, how to entering learning mode.

Continuous on/off of external mechanical switches for 8 times within 5 seconds

The relay cycles turn on for 1 second and turn off for 1 second, the switch status will be uploaded synchronously, that means enter into the learning mode.

2.9.3 Exit learning mode

Received controller command from an unpaired remote control, successfully learned the device. The send the command of exit learning mode.

Received controller command from paired remote control, the device ignored the pairing, exit learning mode.

Timeout: The device did not receive the required RF433 remote control command within 10 seconds, exiting learning mode

Power off: The learning mode failure, the device will be exit learning mode.

Forced return: Short press the external switch or remote control, the learning mode failure, the device

will be exit learning mode.

Less or more than 8 times for turning on/off external switches in 5 seconds, the device will not enter into the learning mode.

2.9.4 Delete the code of remote controller from device

Continuously turn on and off external switches 12 times within 10 seconds, the code will be deleted

3. Electrical Characteristic

3.1 Basic parameters

Test condition: 3.3V±10%, GND=0V ; Temperature 25°C	
Model	TI602B2S
Interface	PWM/IC/GPIO
Working Voltage	3.0V -3.3V
Working Current	Average Current: ≈45mA Max Working Current:350mA
Working temperature	0°C~45°C
Storage environment	Temp: -10°C~+75°C relative humidity 20%RH~80%RH
Wireless network type	WIFI:STA/AP/STA+AP Bluetooth: BLE 5.0
Security	WPS WEP WPA/WPA2 Personal /WPA2 Enterprise /WPA3
Encryption type	WEP64/WEP128/TKIP/AES 128/192/256
Firmware update	OTA Remote upgrade

3.2 WIFI parameters

Test condition: 3.3V±10%, GND=0V ; Temperature 25°C	
WiFi Standard	IEEE 802.11b/g/n
Frequency Range	WIFI:2412-2482MHz BT:2.400GHz to 2.4835GHz
Working Voltage	3.0V -3.3V
Transmitting Power	802.11b:±19dBm(@11Mbps) 802.11g:±16dBm(@54Mbps) 802.11n:±16dBmReceiving tivi(@HT20MCS7)
Receiving Sensitivity	802.11b: -92 dBm(@11Mbps, CCK) 802.11g: -76 dBm(@54Mbps, OFDM) 802.11n: -72 dBm)MCS7)
Antenna Type	PCB antenna on board

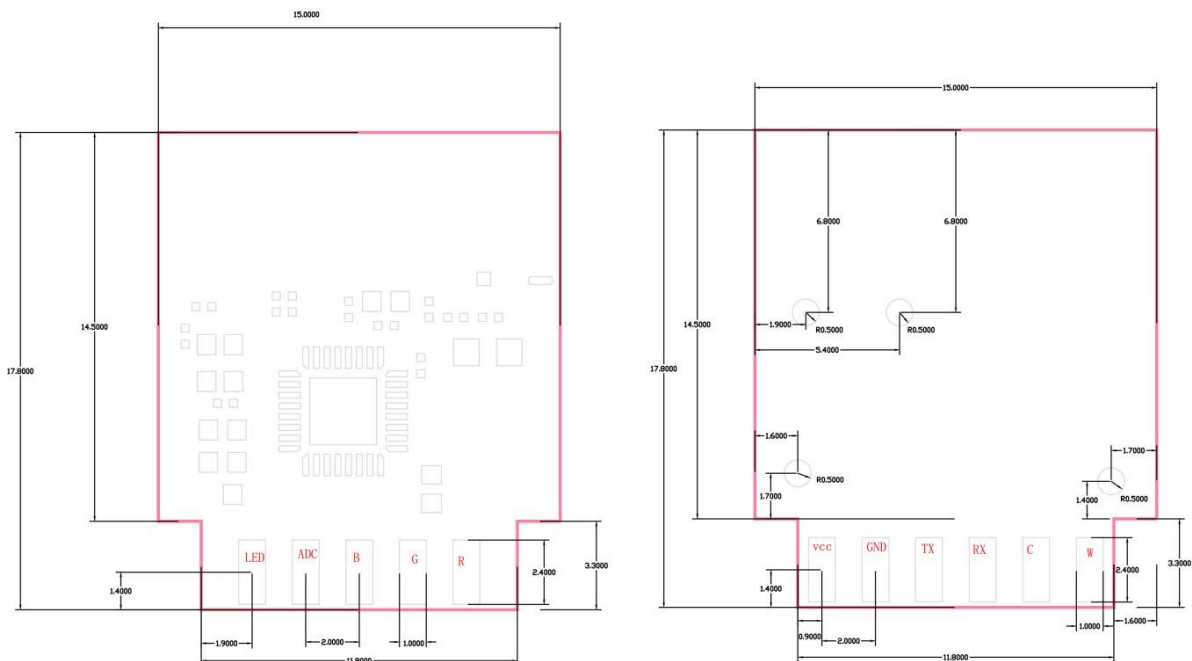
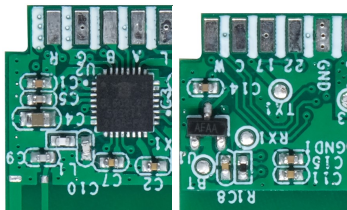
3.3 WIFI RF parameter

Test condition: 3.3V±10%, GND=0V ; Temperature 25°C				
Item	Min	Typ	Max	Unit
Input frequency	2412	-	2482	MHz
Output impedance	-	50	-	Ω
Input reflection	-	-	-10	dB

3.4 Sensitivity

Item	Min	Typ	Max	Unit
CCK 1Mbps		-96		dBm
CCK 11Mbps		-90		dBm
6Mbps(1/2BPSK)		-92		dBm
54Mbps(3/4 64-QAM)		-76		dBm
HT20, MCS7(65Mbps, 72.2Mbps)		-72		dBm

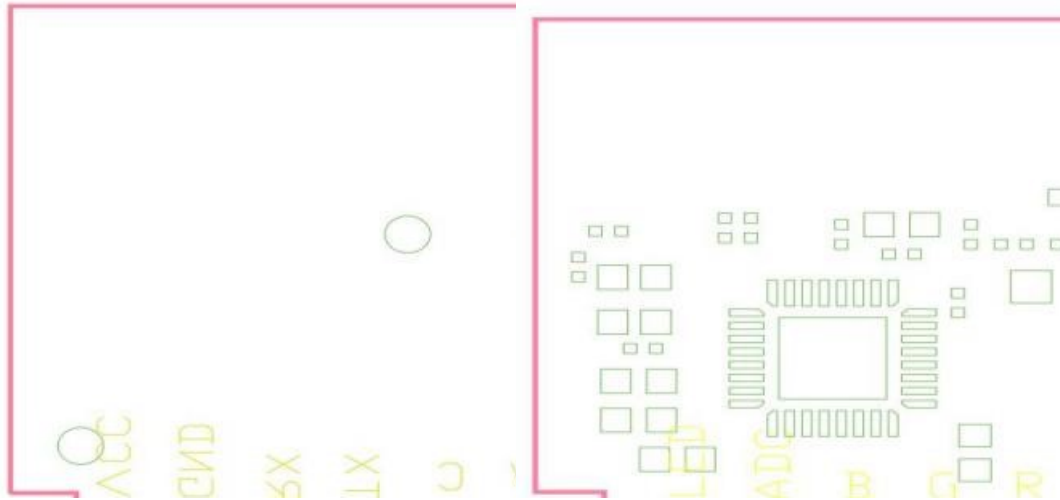
4. Module Size



Dimension unit: MM Tolerance: ±0.1MM

Module size:17.8mm(L)*15mm(W)*2.6mm(D)

5. Pin Assignment



Pin	Symbol	Function
1	VCC	DC Power
2	GND	GND
3	IO22	The impact of mechanical button input on the device
4	IO17	NC
5	C	433 signal input
6	W	WiFi indicator light
7	R	Relay control, Active High level
8	G	Button input, Active GND. External needs to be pulled up. Button control the relay pin flipping, long press 5s enter into pairing mode
9	B	NC
10	ADC	NC
11	LED	NC

6. Reference design for PCBA

Attention of PCB layout and Module

1. When laying out a PCB, pay attention to the placement of the modules and try to stay as far away from interference sources as possible: magnetic components (such as motors, inductors, transformers, etc.), high-frequency signal devices (such as crystal oscillators, high-frequency clock signals, etc.).
2. Clear space is required in the module PCB antenna area and the 15mm external expansion area (copper laying, wiring, and component placement are strictly prohibited)
3. The capacitance of the power supply (VCC) pin of the module and the capacitance and resistance of other pins of the module should be placed as close as possible to the module pins, and the wiring path should be short.

7. Environment about Working and Storage

Working Temp	Temperature:-20°Cto+85°C
	Relative Humidity:10-90%(non-condensing)
Storage Temp	Temperature:-40°Cto+85°C(non-condensing)
	Relative Humidity:10-90%(non-condensing)

8. Package details

- Packed as vacuum reel or tray (TDB)
- Reel color: blue Carton size: 33.7cm*33.7cm*9.5cm
- 2 packs desiccant, 1 pcs humidity card

